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Pothole detection and International Roughness Index (IRI) calculation using ATVs for road monitoring

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The accelerated deterioration of roads is conditioned by parameters such as climate change, poor construction, and heavy vehicle traffic. Two relevant measures to monitor the condition of a road are the International Roughness Index (IRI) and the number of functional failures in a segment, mainly potholes, since they are associated with higher risks such as accidents or damage to vehicle mechanics. In the state of the art, pothole detection or International Roughness Index (IRI) calculation algorithms are proposed, but they use vehicles designed to produce less vibration and use phones that decrease the performance of the embedded sensors. In addition, some works propose complex algorithms of higher computational load that leads to use more hardware and power consumption. In this context, the present work aims to monitor the condition of a road through low-cost dedicated sensors implemented in an urban patrolling all-terrain vehicles (ATVs), where energy consumption is optimized using low-complexity signal processing techniques for noise reduction and detection algorithms. The results show an average accuracy of 90.5% in the detection of potholes, a relative error of 8.41% in the calculation of the International Roughness Index (IRI) and an average reduction of 65.4% in the monitoring time.

The construction of roads makes it possible to communicate different cities, which improves the development of a population. These constructions must be monitored due to wear and tear and the generation of functional failures. These failures arise due to different factors such as vehicle load, vehicle weight, weather conditions, etc.[1]. There are several functional failures but the most relevant one is the pothole, which affects driver comfort, can induce accidents, and affect vehicle mechanics[2]. On the other hand, one of the parameters that allows quantifying the comfort and condition of a road is the International Roughness Index (IRI).This measure is obtained by accumulating the vertical displacements measured with different sensors along a road segment, where there may be potholes, cracks, etc., resulting in measurements in units of m/km. A low IRI value indicates greater comfort for users and better pavement condition. Roadway monitoring is performed through different types of sensors, but in particular automatic monitoring is executed with systems that employ LiDAR (Light Detection and Ranging) sensors or RGB-D (Red Green Blue and Depth) cameras. These allow collecting data in three dimensional form and with maximum accuracy[3]. However, being inaccessible due to the high cost, it opens a gap for the development of new techniques and the use of low-cost sensors to achieve similar results. Another sensor traditionally used in bump detection is the accelerometers embedded in phones. However, the phone limits the scalability of monitoring in different positions for a robust and redundant measurement. In addition, phones having different processes running in the background limit resources, have high power consumption, get a low sampling rate, and can induce noise from events such as notifications or incoming calls. On the other hand, the traditional use of monitoring is through vehicles, such as cars, vans, etc., which are focused on comfort and stability in driving, which produces a low noise level. In addition, the size of these vehicles makes it difficult to move in confined spaces.

In this context, the proposed system aims to provide a low-cost solution and monitoring of the condition of roads in cities, through an algorithm that detects potholes reliably and the calculation of the IRI, focused on

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the use of an ATV (All-terrain vehicle) type vehicle that provides greater accessibility as a monitoring vehicle and access to reduced roads. The design of the algorithm allows to reduce the vibration noise induced by the ATV and improve the captured signals, thus optimizing the detection of potholes. On the other hand, the use of dedicated accelerometers provides a higher level of samples in one capture, ease of conditioning and scalability for implementation at multiple points.

Although the methods found in the literature (see Table 1) work for the detection of potholes, they mostly use sensors integrated in phones that limit the sampling frequency, which reduces the information collected. The use of integrated sensors in non-dedicated equipment does not provide security in data acquisition, due to the fact that applications are running in the background that can interrupt the acquisition through notifications or alerts. In addition, the use of phones increases the cost of a continuous monitoring system because the hardware contains sensors and electronics that are not useful for the application. On the other hand, the solutions observed in the literature are limited to the use of vehicles of a structure and weight that focus on stability and low noise level, which does not allow its use in vehicles of lower cost, smaller size and weight such as ATVs.

Proposed system

The work proposes a pothole monitoring and IRI calculation that is composed of two ICMP-42688-P modules with accelerometers and gyroscopes sensors. These sensors are implemented in an ATV together with Raspberry Pi3B+ that works as a collection and processing node, allowing to calculate the IRI and extract relevant information for the use of machine learning algorithms in the detection of potholes. The methodology of the system is detailed in Fig. 1.

References	Years	Approach	Strengths	Weaknesses
Varona et al.[4], Kumar et al.[5], Hadj-Attou et al.[6] and Kempaiah et al.[7]	2020–2023	They use techniques such as recurrent networks, 1D-CNN and multilayer perceptron. In addition, some employ methods to increase the amount of data artificially	They collect data on different road types and analyze the best sensor location inside the vehicle	They propose algorithms with high computational load, the use of telephones limits the sampling rate of the integrated sensors and the solution is restricted to the use of private transport vehicles
Wu et al.[8], Oche et al.[9] and Kandoi et al.[10]	2020–2021	They implement Random Forest (RF) algorithms together with processing techniques such as resampling, filtering, segmentation and time domain feature extraction	They collect data on four different models of private transport vehicles and some collect people's opinions on the condition of a road	The use of integrated sensors in phones reduces the sampling rate and scalability. Some work relies on internet connectivity for data storage. In addition, use is restricted to private transport vehicles
Kuthyar et al.[11] and Singh et al.[12]	2020–2023	They employ deep learning models for time series and signal processing techniques through thresholding	They implement texture extraction techniques in time series data, including an application for data management	Requires internet connection for data transmission. Limited to high-end smartphones with high processing capabilities
Lekshmipathy et al.[13] and Shravanth et al.[14]	2021	Bump detection is performed using the threshold method and filtering techniques are applied to enhance the captured signal	They use low computational load algorithms and implement interfaces to visualize the detection points	The implementation of sensors is restricted to the use of private transportation vehicles and the use of sensors embedded in phones reduces the sampling rate and scalability of a solution
Kamalesh et al.[15] and Bansal et al.[16]	2020–2021	They use Random Forest and fixed threshold algorithms. They perform missing data cleaning processes and outlier analysis	They implement dedicated acceleration sensors, gyroscope and localization modules	Low reliability of the detection algorithm and lack of information on the algorithm stages. Limited to the use of private vehicles
Arce-Saenz et al.[17]	2023	Comparisons are made using different machine learning algorithms with features extracted in time and frequency domains	Classifies speed bumps, patches, potholes, or uneven stretches	Internet connectivity is required for signal processing
Mirtabar et al.[18] and Zhang et al.[19]	2022	The quarter-car model (QAVM), filters, integration calculation, and profile elevation are applied for the correlation with the IRI	They collect data using dedicated sensors. They propose a low-cost device. They implement a mobile application for data acquisition and transmission	Dependence on external software for calculating the IRI. Frequent variation of quarter-car model parameters depending on the vehicle
Janani et al.[20]	2021	Power Spectral Density and Welch Periodogram are used for calculating the IRI	Aligns the phone using an external accelerometer sensor	Reduced sampling rate with sensors integrated in phones
Aparna et al.[21], Shaghouri et al.[22] and Kotha et al.[23]	2020–2022	Pothole detection is performed using convolutional networks or deep learning models such as Yolov4 or SSD with mobile net	They implement low-cost solutions such as cameras and sensors integrated into the cell phone or thermal cameras for phones	High computational load, algorithms run on GPU servers. In addition, some perform fixed captures or the accuracy is low, on average 65%
Kumar et al.[24], Kumar et al.[25] , Yuan and Yang.[26]and Ugale et al.[27]	2021–2023	They implement machine learning algorithms with processing methods such as Kalman, Jerks and DWT to improve detection. The gray level concurrency matrix (GLCM) algorithm or neural network model is also used	Dedicated sensors, external server processing and bump and speedbump detection are used. A proposal interprets the signal as an image for crack detection	Is restricted to internet connectivity for cloud processing. Requires higher bandwidth for data transmission to the cloud. In addition, some use high-capacity acquisition equipment

Table 1. Related work.

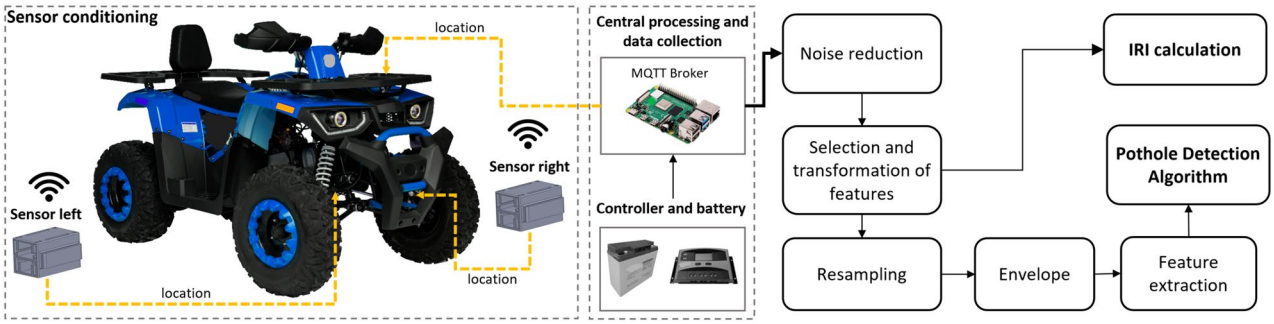


Fig. 1. Diagram of the proposed system.

Sensor conditioning

The sensors are implemented in an ATV, which has the characteristic of light weight and reduced dimensions. This type of vehicle, unlike heavier and more stable vehicles, produces a higher level of unwanted vibration. For the proper implementation of the sensors, the strategic location must be taken into account to balance the sensitivity level of the sensors and a low level of unwanted vibration. For the installation of the sensors, two zones of the ATV, internal and external, were analyzed. Through the visualization of the captured signals, it was observed that the internal structure that joins the tire axle and the steering mechanics, provide greater sensitivity to vibrations and better stability due to the weight driven by the mechanics. The sensors were implemented on each side of the bar, which allows obtaining data from the left and right section of the ATV. Each sensor assembly was secured using pipe clamps. In this work, TDK-InvenSense ICM-42688-P motion sensors are used, which are connected to the ESP32 acquisition and communication module. The two electronic devices are conditioned in a 3D printing enclosure, so that the enclosure can be attached to the ATV structure. Figure 2 shows the conditioning of the different electronic elements, (a) sensor on the right side, (b) with the devices: (1) WiFi modem, (2) Raspberrry Pi 3B+, (3) voltage regulators and (4) charge controller and (c) sensor on the left side.

Data transmission and storage

Data transmission is carried out over a wireless WiFi network using the MQTT(Message Queuing Telemetry Transport) protocol. The MQTT protocol allows low latency in data transmissions and low bandwidth used. The publication structure using the MQTT protocol is done with two topics, 'sensor left' and 'sensor right', each one includes information from the 3 accelerometer axes and 3 gyroscope axes.The Raspberry Pi 4 processor is configured as the "Broker" to manage the published data and through the Node-RED tool installed in the Raspberry an algorithm is executed to process and store the information in a CSV format file. These acceleration and gyroscope data are defined as $a_{t,p}$ and $g_{t,p}$ respectively, where t indicates the axis of each sensor (x, y, z), and p the position of the sensors, left (l) or right (r). The node diagram used by the Node-RED tool and file storage is shown in Fig. 3. It should be noted that the configuration and initialization script for MQTT communication are automatic and are implemented in such a way that the entire process can be initialized by simply turning on the electronic devices.

Data processing

Noise reduction

Although the installation of the sensors in strategic areas reduces the level of unwanted vibration, they still induce a level of noise due to the vibration of the ATV (see Fig. 4). This condition decreases the clarity of the patterns of a bumpy signal. To reduce the noise level at each sensor, noise reduction techniques such as subband decomposition and moving average filtering are implemented to better identify the patterns of the occurrence of a bump and smooth out unwanted variations for the calculation of the IRI index.

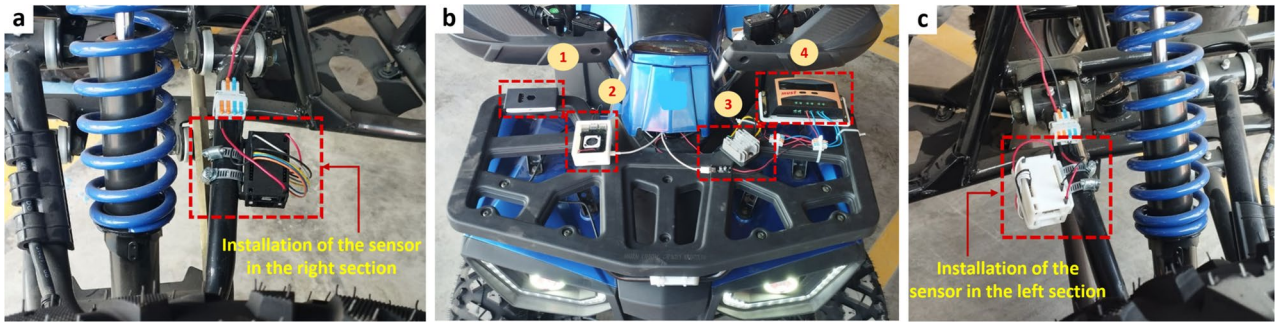


Fig. 2. Implementation of sensors on the ATV.

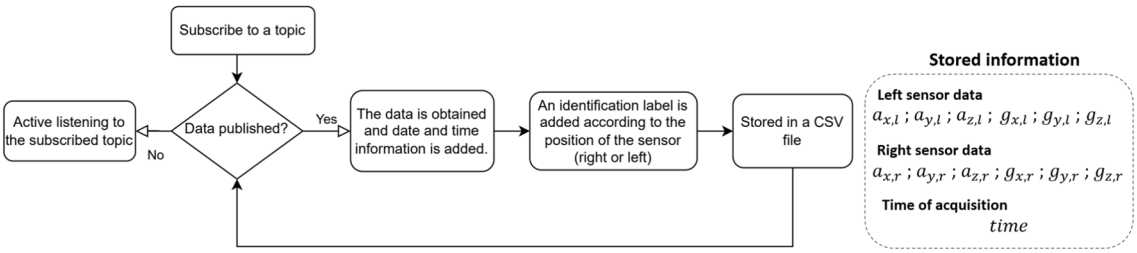


Fig. 3. Flowchart of the algorithm for storing the collected data.

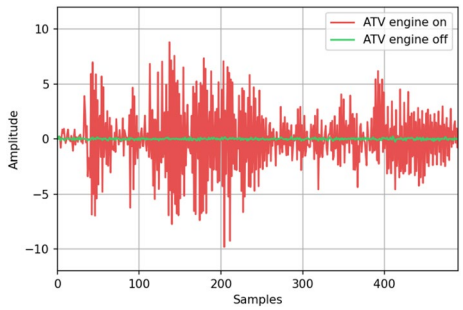


Fig. 4. Vibration generated when the engine is running, and the ATV is parked.

The decomposition of a signal using the wavelet transform allows observing features through applying a type of wavelet filter on the signal and decomposing in a hierarchical way to analyze the signals, which facilitates the elimination of unwanted components. The parameters for its design are based on the level of decomposition, the type of wavelet filter and the selection of the decomposed signals[28]. For the work, a level 3 decomposition with a haar-type wavelet filter was used due to its low computational complexity. In addition, only the approximation and detail components of the last level were selected, this allows eliminating frequencies with a high level of variation or noise and acquiring the signals with better characteristics. To obtain the same length of the input signal, the wavelet transform reconstruction process is performed.

To smooth the signal and obtain a defined signal with a noticeable pattern of amplitude changes, which benefits in bump detection and IRI calculation, the moving average filter is used. This type of filter uses a lower computational load since it is based on averaging a window of samples[29]. A 5-sample size was used for the work since no high grade filtering is needed due to the main filtering process that was performed in the subband decomposition. Figure 5 shows an original signal and the result of applying the noise reduction process. The result of applying the noise reduction method to the acceleration signals $a_{t,p}$ and for gyro signals $g_{t,p}$ is represented as $A_{t,p}$ for the and $G_{t,p}$ respectively.

Feature selection and transformation

For the selection of the variables, we took into account what was mentioned in[30], where the magnitudes of the acceleration data are calculated, since each axis contains relevant information and is grouped into a single variable to reduce the dimensionality of the information. The calculation of the magnitude of the sensor located on the right side uses the accelerations $A_{x,l}$, $A_{y,l}$ and $A_{z,l}$; for the left side $A_{x,r}$, $A_{y,r}$ and $A_{z,r}$ are used. The representation of the magnitude calculations is described in Eqs. (1) and (2), obtaining as output the magnitude $|A_l|$ and $|A_r|$,

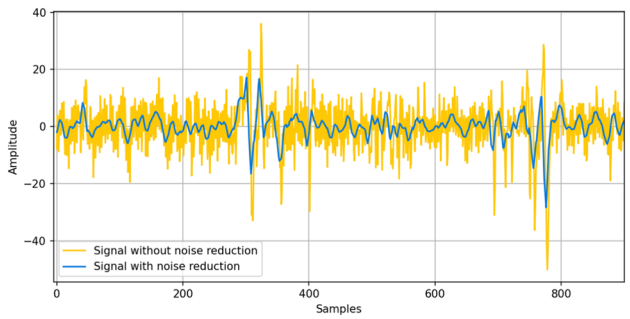


Fig. 5. Result of noise reduction by applying mobile transform and wavelet transform.

right and left respectively. Another variable chosen for the development of the proposed algorithms is the signal of the gyroscopes with x axis since by means of an analysis it is observed a greater presence of patterns in the detection of bumps within this axis of the gyroscope, defined as $G_{x,l}$ and $G_{x,r}$ for the right and left side respectively.

$$|A_l| = \sqrt{(A_{x,l})^2 + (A_{y,l})^2 + (A_{z,l})^2} \tag{1}$$

$$|A_r| = \sqrt{(A_{x,r})^2 + (A_{y,r})^2 + (A_{z,r})^2} \tag{2}$$

Resampling

Acceleration and gyroscope sensors have a sampling frequency that allows knowing the amount of data obtained per second (Hz). Although the sensors have a fixed frequency detailed in their specification documentation, these may vary due to the computational complexity of the algorithms, the type of programming language and the hardware used[31]. In this work the sensors have a sampling frequency of 100 Hz which is detailed in their technical documentation, but at the end of the processes a variable sampling frequency greater than 93 Hz is obtained. Therefore, the resampling method is used, which allows decreasing or increasing the data of a signal to adapt it to a sampling value and achieve a signal analysis with fixed parameters. Within this method, a linear interpolation with low computational complexity is used to obtain new samples and establish a sampling frequency of 100 Hz. Figure 6 shows the interpolation process applied and the samples generated in a 1 s. The result of the resampling to the different signals are defined as $|A_l|^S$, $|A_r|^S$, $G^S_{x,l}$ and $G^S_{x,r}$.

Envelope

The envelope of a signal is useful for decoding modulated digital signals as in the processing of temporal signals that exhibit a high level of variability[32]. The use of the envelope allows to obtain points that write to a signal based on the shape of its amplitude in time, helping to extract features easily and quickly since its amplitude is only positive and decreases its processing in the calculation. For the implementation in this work, the envelope described in Eq. (3) was used, where the signals resulting from resampling are used as *Input* and $h_1(n)$ is a low-pass filter of order 10. The result of the envelope are positive data (see Fig. 7) and reveals more general details for feature extraction, these results are represented as A^E_l, A^E_r, $G^E_{x,l}$ and $G^E_{x,r}$.

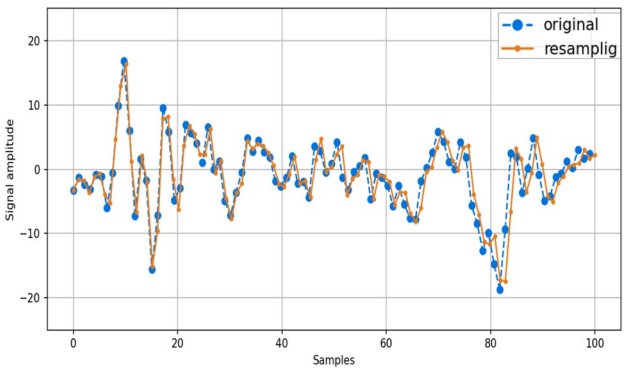


Fig. 6. Signal resampling at 100 Hz.

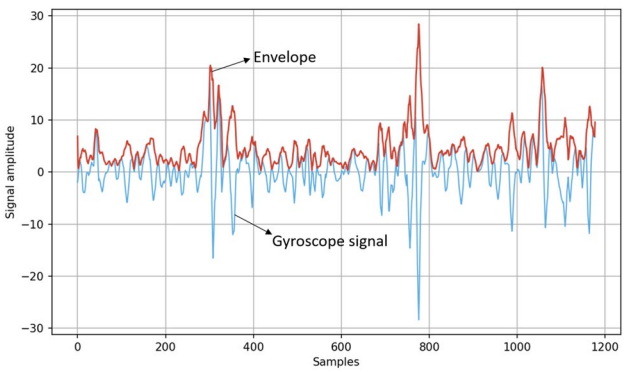


Fig. 7. Result of the envelope of a $G^S_{x,l}$ signal.

$$A_I^E = Input * h_1(n) \tag{3}$$

Labeling and segmentation

The labeling process is performed by selecting a sample window and assigning a class. The selection of the number of samples is based on the analysis of the vehicle speed and the sampling frequency. The higher the sampling rate, the better resolution signals are obtained in less time, allowing a clearer identification of potholes. In this work we use a sampling frequency of 100 Hz and an average speed of 10 km/h, therefore every second the vehicle travels a distance of 27.7 cm with 100 acquired samples. Therefore, the minimum length that can be captured is 0.27 cm. Considering that the bumps observed in multiple segments are larger than 10 cm (see Fig. 8), a fixed window of 30 samples is chosen.

Once the number of samples in a window has been defined, the labeling process is carried out with the support of an Intel DR435 camera located on the front of the ATV. The camera allows capturing RGB images and building a video where to each image is placed the capture time. This helps to identify the time when the ATV passes over the bump and to quickly find the sample window related to the four signals resulting from the envelope process. Once the window where the pothole occurs is identified, it is labeled with a value of '1' and with a value of '0' for windows where there is no pothole. The labels made for the present work are 117 of class "pothole" and 132 of class "no pothole". Figure 9 shows a frame of the video that is related to the window of one of the captured signals.

Pothole detection

Classification algorithms

Classification algorithms can be implemented using basic signal processing rules that apply methods with decision thresholds or using machine learning algorithms. The latter allows robust classification of data through mathematical and statistical calculations[33,34]. Therefore, in this work, supervised learning algorithms are used, the most relevant ones being Support Vector Machine (SVM), K-nearest neighbors (KNN) and Random Forest (RF) algorithms, as highlighted in previous research[5,6,8,15].

The SVM algorithm is a discriminative method that classifies the data and is based on finding an optimal separation hyperplane that allows the training classes to be divided by a maximum distance. In addition, it uses a kernel function that allows to transform the samples to a larger space and to improve the classification[35]. The most commonly used or common types of kernel functions are polynomial, radial basis function (RBF), linear and sigmoidal. The KNN algorithm is a classification and regression method that is based on the calculation of the distance with respect to the K near neighbors, this is obtained by different types of measurement such as Euclidean distance, Manhattan distance or Hamming distance[36]. In addition, the selection of the optimal K is important, since a too high or too low value causes over-fitting or increased noise respectively. The RF algorithm is a supervised method that groups multiple random decision trees, where its learning method as a whole is

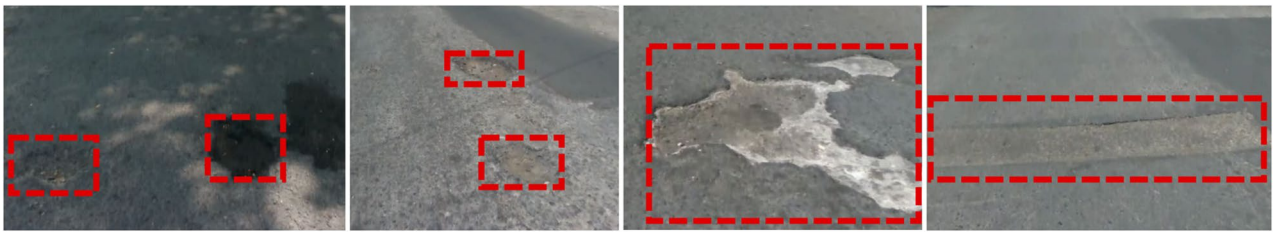


Fig. 8. Types of potholes identified in the capture.

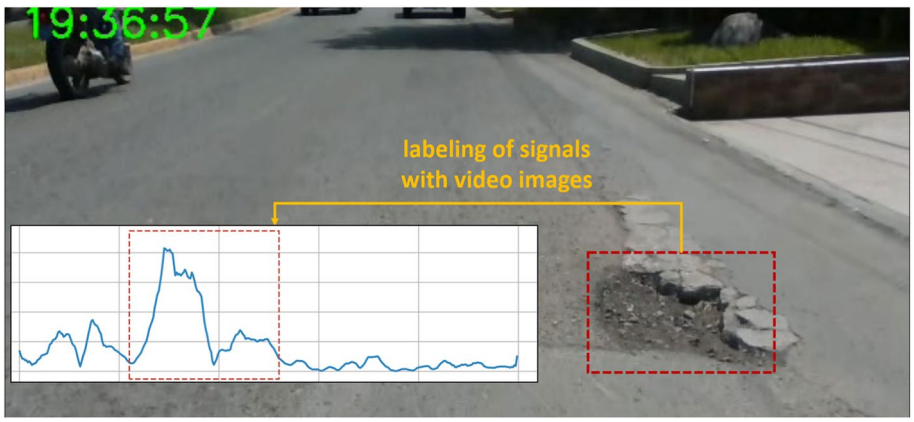


Fig. 9. Camera-assisted identification of pothole signals.

based on "Bagging", allowing to perform parallel training to a subset of random data[37]. In[38] they mention two relevant parameters to implement RF, the number of decision trees and the number of variables to be selected.

Feature extraction
For the inputs of the machine learning algorithms, characteristics of the samples that compose each labeled window are obtained. In this work we used time and frequency domain measures to serve as representative characteristics that improve the training of the machine learning models. In the time domain, measures were chosen that represent significant information, such as minimum, maximum, average and standard deviation, which in several works such as[3–5] use these measures and achieve important results. On the other hand, features in the frequency domain are extracted based on applying a fast Fourier transform and through the result, measures such as maximum, average, and standard deviation are obtained.

Metrics for classification models
To evaluate the performance and selection of the classification algorithm, confusion matrices and Accuracy, Precision, Recall, F1-Score metrics are used. In addition, the algorithms are evaluated by means of the ROC measure, where the specificity and sensitivity metrics are used. These metrics are defined in Table 2.

IRI calculation

The computational calculation of IRI (International Roughness Index) is estimated by measuring a longitudinal pavement profile, this is done using data collected from accelerometer and gyroscope sensors, obtaining the vertical variations. The authors of[18–20] use a simplified equation of the general quarter car equation, allowing to estimate the IRI_p, described in Eq. (4), index with greater ease and simplicity, where p indicates the position of the installed sensor, right (r) or left (l) and tr indicates the distance of the monitored section, t_{start} and t_{stop} indicate the time when the capture starts and ends respectively. In addition, in[19] they detail that in order to obtain a better reference of the vertical acceleration, the process of obtaining the average of each acceleration axis $\overline{A_p} = (\overline{A_{x,p}}, \overline{A_{y,p}}, \overline{A_{z,p}})$ during 5 seconds with the vehicle parked is performed. This procedure allows calculating the vertical acceleration, detailed in Eq. (5), in a more representative way since the sensors are not perfectly aligned.

$$IRI_p = \frac{\int\int_{t_{start}}^{t_{stop}} |av_p|(dt)^2}{tr} \tag{4}$$

$$av_p = \frac{A_p \cdot \overline{A_p}}{|\overline{A_p}|} = A_{x,l} * \overline{A_{x,l}} + A_{y,l} * \overline{A_{y,l}} + A_{z,l} * \overline{A_{z,l}} \tag{5}$$

It should be noted that in this work a sensor is used for each side of the ATV, therefore the final calculation of the IRI is estimated by averaging (IRI_m), as detailed in the following Eq. (6).

$$IRI_m = \frac{IRI_l + IRI_r}{2} \tag{6}$$

Test scenarios

For the present work, tests were carried out in two scenarios. These are located in the district of Santiago de Surco, Lima, Peru. The two scenarios are of flexible asphalt pavement type, are two-lane and contain different faults, but mainly potholes. In the first scenario (see Fig. 9a) approximately 4 km are monitored and in the second scenario 1.4 km (see Fig. 9b). In each scenario an average speed of 10 km/h was set and as a monitoring protocol a second pass was established, one to know the pothole points and the second to be able to pass over the most relevant potholes. It should be noted that in these scenarios an approximate time of day was chosen where traffic is low.

Results

Evaluation of algorithm

Evaluation in scenario 1
Based on the different supervised classification algorithms mentioned in section 3.6, a training was performed with data from only scenario 1 where the training and validation data split is 70% and 30% respectively. The results of the metrics obtained for each classification algorithm are shown in Table 3, where the initial parameters established in each algorithm are detailed. In addition, as part of the choice of the classification algorithm, a comparison of the ROC curve was performed to observe the discriminatory capacity of each algorithm. From the metrics obtained, it is observed that the SVM algorithm performs better than the other algorithms, with an

Metrics	Accuracy	Precision	Recall	F1	Specificity	Sensitivity
Equation	$\frac{TP+TN}{TP+FP+TN+FN}$	$\frac{TP}{TP+FP}$	$\frac{TP}{TP+FN}$	$\frac{2 \times Presicion \times Recall}{Presicion \times Recall}$	$\frac{TN}{TN+FP}$	$\frac{TP}{TP+FN}$

Table 2. Performance evaluation metrics.



(a) Scenario 1. **(b)** Scenario 2.

Fig. 10. Scenarios for the evaluation of the proposed system. The base map is from Google Earth (https://earth.google.com/), Airbus, June, 2024.

Model	Accuracy	Precision	Recall	F1-score	AUC
SVM	0.91	0.92	0.91	0.91	0.917
KNN	0.73	0.74	0.73	0.73	0.733
RF	0.73	0.74	0.73	0.73	0.733
Linear regression	0.55	0.59	0.55	0.52	0.567
Decision tree	0.64	0.67	0.64	0.63	0.650

Table 3. Results of the metrics evaluated.

accuracy of 92% and a ROC curve with Area under curve (AUC) of 0.917. The outstanding results of the SVM algorithm allow it to be chosen as the classification algorithm, establishing the use of kernel of type 'sigmoid'.

Evaluation in scenario 2

To measure the performance of the classification algorithm, the algorithm chosen in the previous evaluation was used with data from scenario 1. In this evaluation process, the algorithm trained to examine scenario 2 described in section **Test scenarios** was used. The results of the pothole detection are shown in Fig. 11, which visually details the detections and non detections in 100 m segments, denoted S1 to S14, over a 1.4 km path. It should be noted that each quadrant on the "travel distance" axis represents 2 m. The evaluation in this scenario results in an accuracy of 90.5% on average, which represents a relevant level of detection considering the noise induced by the ATV.

IRI evaluation

The evaluation of the IRI is performed by comparing the proposal of this work and the result collected in a pay phone application "ROADROID" which is focused on the monitoring of IRI through the coupling of the phone in the internal part of a vehicle. It is worth mentioning that the tests performed with the "ROADROID" application were carried out from a car model "GLORY 500" of make "DFSK". In addition, the resulting data are published on their website (https://www.roadroid.com/map), but in a simplified mode.

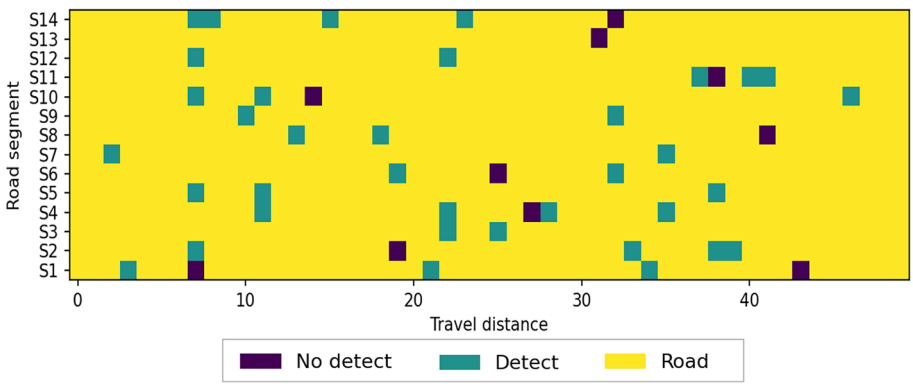


Fig. 11. Segment detection of potholes in scenario 2.

For this evaluation, scenario 1 was used, monitoring only the right lane for 20 segments, where each segment represents 44 m. Figure 12 shows the data resulting from the calculation performed in this work and the results of the ROADROID application. The relative error of the calculation averages 8.41%.

Operating performance

To evaluate the operational performance of the proposal, a comparison was made between a daily process of manual detection of potholes and the automatic detection of the proposal of this work. This evaluation is performed for scenario 2 where the manual process is performed with annotations of the potholes found to finally count them by segments of 100 m. In this manual process the time for each segment is measured by a telephone timer, where mainly the number of potholes is directly related to the increase in time. In addition, physical wear and tear on the observer causes measurement times to vary.

The automatic supervision process allowed to use on average 65.4% less time compared to the manual process (see Fig. 13). In addition, as part of the performance, the energy consumption in Wh is evaluated, for this process a current and voltage monitoring is performed at an interval of 1s during 60 min. From the measurements, an average voltage of 5.0982 V and an average current of 0.5897 A are obtained, consuming an average of 3.065 Wh. Considering that the typical average consumption of a cell phone alone is 5 Wh[39], the energy consumption is reduced by 38.7%.

Conclusion

The proposal focuses on detecting potholes and calculating the IRI through the use of dedicated and low-cost sensors that are implemented in an ATV. In addition, the detection time and power consumption are optimized through low complexity techniques. The results obtained from the experiment are relevant, since an accuracy of 90.5% is obtained in the detection of potholes and a relative error of 8.41% in the calculation of the IRI. In addition, from the comparison between the manual process of pothole detection and the proposed automatic process, the time is optimized with a reduction of 65.4%. This allows generating a preliminary report of the condition of a road. In addition, the proposed automatic process increases the safety of people, since the manual process requires walking on the road to verify the pothole detection. Finally, the use of dedicated sensors allows a higher resolution in the signals captured for pothole detection. Also, by implementing dedicated devices these have the advantage of portability and scalability. From the performance tests, an average consumption of 3065 Wh is obtained, decreasing the average consumption of a phone by 61.3%.

For future work, it is intended to condition other types of sensors such as RGB cameras or acoustic sensors that through optimized neural network algorithms allow classifying a greater number of types of faults. The combination of different types of sensors would provide reliability in detection and obtain characteristics such as size and depth. Also, as future works, the minimization of the size of the sensing devices could be emphasized and a web application could be made available for the consultation of points referred to the detection of faults.

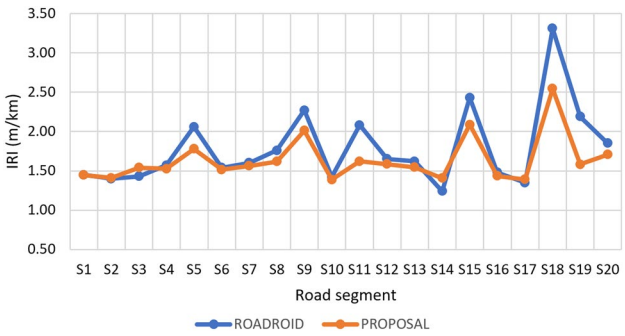


Fig. 12. Comparison of the estimated IRI and ROADROID.

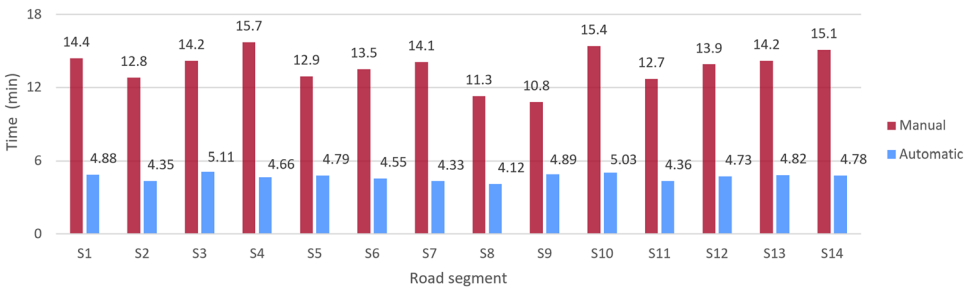


Fig. 13. Time comparison between manual and automatic monitoring for each segment.

Data availability

The data collected for the study has been deposited on Github with the following link: https://github.com/KevinGuerraHuaman/road_sensor_dataset.

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Conceptualization, K.G. and C.R.; methodology, K.G., C.R. and G.Z.; software, K.G. and M.S.; validation, K.G., C.R. and M.S.; formal analysis, K.G.; investigation, K.G, M.S. and J.M.; data curation, K.G and M.S.; writing-review and editing, K.G, C.R., M.S., G.Z and J.M.; project administration, C.R. All authors have read and agreed to the published version of the manuscript.

Competing interests

The authors declare no competing interests.

Additional information

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